Mass conservation for fractional porous-medium flow on noncompact Riemannian manifolds

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Abstract

We prove mass conservation for the spectral fractional porous-medium equation on complete noncompact Riemannian manifolds with Ricci curvature bounded below and a Euclidean-dimensional Faber–Krahn inequality. The result holds for every nonnegative weak dual solution that is spatially integrable at each finite time, for arbitrary nonnegative integrable initial data, all exponents $m > 1$, and all fractional orders $0 < s < 1$. Thus the prescribed weak solution class cannot lose mass at spatial infinity. The proof first obtains a mass upper bound from positivity of the integrated Green potential. A positive energy identity after heat regularization then yields global integrability of the nonlinear power on every strictly positive time interval. A conservative linear resolvent converts this integrability into mass equality at every finite time.

1 Introduction

Let M be a complete connected noncompact Riemannian manifold. We consider the fractional porous-medium equation

$$\partial_t u + (-\Delta_M)^s(u^m) = 0, \qquad m > 1, \quad 0 < s < 1, \tag{1}$$

where the fractional power is defined spectrally. For a nonnegative density, preservation of its integral expresses the absence of mass loss at spatial infinity. On a noncompact manifold this conclusion requires an argument even when the underlying linear diffusion preserves mass: the weak formulation permits only compactly supported tests, and the nonlinear power need not initially be globally integrable.

Berchio, Bonforte, Grillo and Muratori developed a weak dual solution theory for (1) under a Ricci lower bound and a uniform Faber–Krahn inequality [1]. Their formulation evolves the Green potential of u, and accommodates a weighted L^1 continuity condition. Their existence and smoothing results provide solutions obtained through nonlinear semigroup constructions and monotone approximation. Conservation of mass is listed among the open questions in Section 7 of that work. An estimate established for the constructed solutions cannot simply be assigned to every solution satisfying the weak dual identity.

For manifolds with nonnegative Ricci curvature and a finite fractional Green kernel, von Criegern, Grillo and Monticelli [2, Corollary 8.6] prove mass conservation for weak dual solutions obtained by monotone approximation from solutions associated with bounded integrable data. The present theorem treats any spatially integrable weak dual solution in the stated class and allows a general Ricci lower bound.

Here we prove the following assertion. Suppose M has dimension $N \geq 2$, its Ricci curvature is bounded below, and its first Dirichlet eigenvalue obeys

$$\lambda_1(D) \geq c\ \mathrm{Vol}(D)^{-2/N}, \qquad c > 0,$$

for every relatively compact smooth domain D. Then every nonnegative weak dual solution of (1), with nonnegative initial data in $L^1(M)$ and with $u(t) \in L^1(M)$ at each finite time, satisfies

$$\int_M u(t)\,d\mu = \int_M u(0)\,d\mu \qquad \text{for every } t > 0.$$

The exact weighted solution definition and the full statement are given in Section 2. In particular the conclusion concerns all solutions in that class; it does not require identifying them with a particular construction.

The dual evolution says that the Green potential decreases by the time integral of u^m. That time integral is initially controlled only locally. We first show that a nonnegative Green potential of an integrable signed function forces its integral to be nonnegative. The proof uses integrable upper-level truncations of the potential and a conservative linear resolvent. This gives a mass upper bound for the given weak dual solution.

The remaining issue is global integrability of u^m. Heat regularization makes the potential energy finite, even when the initial unregularized energy is infinite. Symmetry of the Green kernel gives an exact identity involving only nonnegative terms and endpoint values. Summing it over time partitions yields a dissipation inequality without differentiating the solution or assuming energy regularity. A Green-potential estimate and interpolation make the energy bound uniform at positive times. Once u^m is globally integrable on positive time intervals, the same resolvent argument gives equality of endpoint masses.

The result addresses conservation in the stated spatially integrable class; it does not assert uniqueness in the larger weighted solution space. Section 2 records the linear estimates and normalization. Section 3 proves the potential-order argument, and Section 4 proves the energy estimate and mass conservation. All integrals use Riemannian volume μ. Operator powers are spectral, and C denotes a positive constant depending on the fixed geometric and equation parameters; its value may change between formulas.

2 Setting and linear estimates

Throughout, M has dimension $N \geq 2$ and satisfies

$$\mathrm{Ric} \geq -(N-1)k\,g, \qquad k \geq 0, \qquad \lambda_1(D) \geq c\,\mu(D)^{-2/N}, \quad c > 0, \tag{2}$$

on every relatively compact smooth domain D. The first eigenvalue is understood variationally, as the infimum of $\int |\nabla\phi|^2 / \int |\phi|^2$ over nonzero $\phi \in C_c^\infty(D)$.

The same inequality holds on every relatively compact open set Ω. Indeed, for each $\phi \in C_c^\infty(\Omega)$ choose a smooth compactly supported function in Ω equal to one near $\operatorname{supp}\phi$. A regular superlevel set between zero and one is a relatively compact smooth domain containing this support; its existence follows from Sard's theorem [4, Theorem 7.2]. If it is disconnected, apply (2) to its components. Since each component has volume at most $\mu(\Omega)$, summing their Rayleigh inequalities and taking the infimum over ϕ gives the assertion with the same c and N. Thus the Faber–Krahn condition in [1, Assumption 2.1] follows from (2).

Write $A = -\Delta_M$ for its nonnegative self-adjoint Friedrichs realization, $P_t = e^{-tA}$, and $L = A^s$. The heat kernel is symmetric, nonnegative, and satisfies

$$p_t(x,y) \leq Ct^{-N/2}\exp[-d(x,y)^2/(Ct)], \qquad t > 0; \tag{3}$$

see [1, (2.3)]. The heat flow is sub-Markov and contracts L^1 and L^∞. Define the normalized Green kernel and its positive integral operator by

$$G_s(x,y) = \frac{1}{\Gamma(s)}\int_0^\infty p_t(x,y)t^{s-1}\,dt, \qquad Gf(x) = \int_M G_s(x,y)f(y)\,d\mu(y). \tag{4}$$

For nonnegative inputs the integral is initially allowed to be infinite. Notice that $N > 2s$ in the entire parameter range.

For $x_0 \in M$, put

$$\|f\|_{1,x_0,G_s} = \int_{B_1(x_0)} |f|\,d\mu + \int_{M\setminus B_1(x_0)} |f(x)|G_s(x,x_0)\,d\mu(x).$$

Definition 2.1. A nonnegative measurable function u is a weak dual solution with initial datum $u_0 \geq 0$ if, for every $T > 0$ and $x_0 \in M$,

$$u \in C([0,T]; L^1_{x_0,G_s}), \qquad u^m \in L^1((0,T); L^1_{\mathrm{loc}}(M)), \qquad u(0) = u_0,$$

and

$$\int_0^T \int_M [Gu\,\partial_t\psi - u^m\psi]\,d\mu\,dt = 0 \tag{5}$$

for every smooth ψ compactly supported in $(0,T)\times M$.

This is the potential formulation of [1, Definition 2.1], with the normalized kernel (4) and with only the compact smooth tests needed here.

Theorem 2.2. *Assume (2), $m > 1$, and $0 < s < 1$. Let $u_0 \geq 0$ belong to $L^1(M)$, and let u be any nonnegative weak dual solution in Definition 2.1. If $u(t) \in L^1(M)$ for every finite $t > 0$, then*

$$\int_M u(t)\,d\mu = \int_M u_0\,d\mu \qquad \text{for every finite } t > 0.$$

We use two linear potential estimates from [1]. First, for each fixed ball $B_\sigma(x_0)$,

$$\|Gf\|_{L^1(B_\sigma(x_0))} \leq C_{\sigma,x_0}\,\|f\|_{1,x_0,G_s}\,; \tag{6}$$

this is the estimate in Remark 2.1, using Lemma 3.2 there. It implies continuity of Gu in L^1_{loc}. Second, Lemma 5.2 gives, for $1 < p < N/(2s)$,

$$\|Gf\|_{pN/(N-2sp)} \leq C_p\,\|f\|_p\,. \tag{7}$$

All source locators here refer to arXiv version 2 of [1]. Its displayed heat potential (2.6) omits $1/\Gamma(s)$. If that printed kernel is denoted by $\widehat{G}_s$, then $\widehat{G}_s = \Gamma(s)G_s$. The source weighted norm and the one above are equivalent, with comparison factors $\min\{1,\Gamma(s)^{-1}\}$ and $\max\{1,\Gamma(s)^{-1}\}$. Dividing the potential estimates by $\Gamma(s)$ gives (6)–(7). We justify the exact spectral representation separately.

Lemma 2.3. *The heat semigroup P_t and the fractional semigroup $S_r = e^{-rL}$ preserve integrals on $L^1(M)$. For nonnegative $f \in L^1(M)$,*

$$Gf = \int_0^\infty S_r f\,dr \tag{8}$$

almost everywhere. If $p_w = N/(N-2s)$, then

$$\mu\{G|f| > \lambda\} \leq C(\|f\|_1/\lambda)^{p_w}, \qquad \lambda > 0, \tag{9}$$

and, for every $\varepsilon > 0$,

$$\|P_\varepsilon G|f|\|_\infty \leq C\varepsilon^{s-N/2}\,\|f\|_1\,, \qquad P_\varepsilon Gf = GP_\varepsilon f \quad (f \geq 0). \tag{10}$$

Proof. Bianchi–Setti's cutoffs [3, Corollary 2.3], with their curvature exponent zero, give $0 \le \chi_R \le 1$ in $C_c^\infty(M)$, equal to one on $B_R(o)$, and $\|A\chi_R\|_\infty \le C/R$, $R \ge 1$. Their curvature constant may be chosen with $\kappa^2 \ge k$; their support parameter is fixed independently of R. Spectral calculus gives

$$P_t\chi_R - \chi_R = -\int_0^t P_\rho A\chi_R \, d\rho.$$

For bounded compact f, symmetry and contraction therefore imply

$$|\langle P_t f - f, \chi_R\rangle| \le t \, \|f\|_1 \, \|A\chi_R\|_\infty \longrightarrow 0.$$

Dominated convergence gives $\int P_t f = \int f$. Density and contraction extend this to all L^1 inputs. In particular $P_t 1 = 1$ in the kernel sense. The subordination formula for $0 < s < 1$ is

$$S_r f = \int_0^\infty P_\rho f \, d\nu_r(\rho), \qquad \int_0^\infty e^{-\lambda\rho} d\nu_r(\rho) = e^{-r\lambda^s};$$

see the linear semigroup discussion in [1, proof of Proposition 5.1]. The measures ν_r have total mass one, by letting $\lambda \downarrow 0$. Tonelli now proves conservation for S_r.

For $\lambda > 0$, the two scalar integrals

$$\frac{1}{\Gamma(s)} \int_0^\infty e^{-t\lambda} t^{s-1} \, dt = \lambda^{-s} = \int_0^\infty e^{-r\lambda^s} \, dr$$

agree. Spectral calculus [5, Section 3.1] applies first to quadratic pairings with bounded compact inputs h. These pairings are finite: splitting the heat integral at one gives

$$\|G|h|\|_\infty \le \frac{1}{\Gamma(s)} \left(\frac{\|h\|_\infty}{s} + \frac{C \, \|h\|_1}{N/2 - s} \right).$$

There is no L^2 zero mode, since (3) and symmetry give $\|P_t h\|_\infty \le C t^{-N/4} \, \|h\|_2$. An invariant L^2 function is therefore zero. Polarization identifies mixed compact pairings. Increasing bounded compact truncations of a nonnegative L^1 input and applying Tonelli gives (8).

To prove (9), split (4) at a time $T > 0$. The short part has L^1 norm at most $CT^s \, \|f\|_1$, and the long part has L^∞ norm at most $CT^{s-N/2} \, \|f\|_1$. For $f \ne 0$ choose T so that the latter bound is $\lambda/2$, and use Chebyshev's inequality on the short part. This gives the exponent p_w in (9). The zero input is immediate. In particular these potentials are finite almost everywhere. Finally, Tonelli and the heat composition law give commutation and

$$\|P_\varepsilon G|f|\|_\infty \le C \, \|f\|_1 \int_0^\infty (\varepsilon + t)^{-N/2} t^{s-1} \, dt \le C' \varepsilon^{s-N/2} \, \|f\|_1 .$$

The last integral is finite because $0 < s < N/2$. □

3 Potential order and endpoint evolution

We first extract information about mass from an ordered potential. The potential itself need not be integrable.

Lemma 3.1. *Let $h \in L^1(M)$ be real and suppose $W = Gh \ge 0$ almost everywhere. Then $\int_M h \, d\mu \ge 0$. If in addition $W \in L^1(M)$, then $\int_M h \, d\mu = 0$.*

Proof. Fix $\alpha > 0$ and define the positive resolvent and its Markov normalization by

$$R_\alpha = \int_0^\infty e^{-\alpha r} S_r \, dr, \qquad T_\alpha = \alpha R_\alpha.$$

Lemma 2.3 implies

$$T_\alpha 1 = 1, \qquad \int_M R_\alpha f \, d\mu = \alpha^{-1} \int_M f \, d\mu, \qquad \int_M T_\alpha f \, d\mu = \int_M f \, d\mu \quad (f \in L^1). \tag{11}$$

The extension to nonnegative measurable functions is by monotone kernel integration. For $f \geq 0$ in L^1, (8) and Tonelli yield

$$T_\alpha G f = \int_0^\infty (1 - e^{-\alpha v}) S_v f \, dv = Gf - R_\alpha f.$$

The coefficient follows by setting $v = r + t$ and integrating $\alpha e^{-\alpha r}$ over $0 < r < v$. All these potentials are finite almost everywhere by (9). Applying the identity to the positive and negative parts of h, dominated by $G|h|$, gives

$$W - T_\alpha W = R_\alpha h =: g \in L^1(M). \tag{12}$$

Since $0 \leq W \leq G|h|$, estimate (9) and $p_w > 1$ show that for each $\ell > 0$,

$$\int_M (W - \ell)_+ \, d\mu = \int_\ell^\infty \mu\{W > \lambda\} \, d\lambda < \infty.$$

Positivity and $T_\alpha 1 = 1$, applied to $W \leq (W - \ell)_+ + \ell$, imply

$$(T_\alpha W - \ell)_+ \leq T_\alpha (W - \ell)_+.$$

Both truncated functions are integrable, so (11) gives

$$\int_M \left[(W - \ell)_+ - (T_\alpha W - \ell)_+ \right] d\mu \geq 0.$$

The absolute value of the integrand is bounded by $|g|$. Letting $\ell \downarrow 0$ in dominated convergence gives $\int g \geq 0$, hence $\int h = \alpha \int g \geq 0$. If $W \in L^1$, direct integration of (12) and (11) gives $\int g = 0$. □

Proposition 3.2. *Every solution in Theorem 2.2 satisfies, for all* $0 \leq a < b < \infty$,

$$Gu(a) - Gu(b) = W_{a,b} := \int_a^b u(t)^m \, dt \geq 0 \quad \text{in } L^1_{\mathrm{loc}}(M) \tag{13}$$

and

$$\|u(b)\|_1 \leq \|u(a)\|_1 \leq M_0 := \|u_0\|_1 . \tag{14}$$

Moreover $u(t) \to u_0$ *strongly in* $L^1(M)$ *as* $t \downarrow 0$.

Proof. The defining weighted continuity implies local L^1 continuity of u, by covering compact sets with finitely many unit balls. Estimate (6) gives the same continuity for Gu. Using product tests in (5) shows that $\partial_t Gu = -u^m$ distributionally. For each compact smooth spatial test, the right side is integrable up to 0 and T. Its indefinite integral is continuous; subtracting it leaves a continuous scalar function with zero distributional derivative, hence a constant. Equality against all spatial tests gives (13) for every endpoint, including zero. In particular no time representative is discarded.

Lemma 3.1 applies to $h = u(a) - u(b)$ and (13), and proves (14). For initial strong continuity, fix $\delta > 0$ and a compact K with $\int_{M \setminus K} u_0 \, d\mu < \delta$. For small $t > 0$, local convergence gives $\int_K |u(t) - u_0| \, d\mu < \delta$, while

$$\int_{M \setminus K} u(t) \, d\mu \leq M_0 - \int_K u(t) \, d\mu \leq \int_{M \setminus K} u_0 \, d\mu + \delta < 2\delta.$$

It follows that $\|u(t) - u_0\|_1 < 4\delta$. □

4 Energy and conservation

If $M_0 = 0$, Proposition 3.2 already proves Theorem 2.2. Hence assume $M_0 > 0$. We regularize with the ordinary heat semigroup, which commutes with the spectral Green operator.

Proposition 4.1. *For $\varepsilon > 0$, define*

$$f_\varepsilon(t) = P_\varepsilon u(t), \qquad V_\varepsilon(t) = G f_\varepsilon(t) = P_\varepsilon G u(t), \qquad E_\varepsilon(t) = \int_M f_\varepsilon(t) V_\varepsilon(t)\, d\mu.$$

These energies are finite, and for every $0 \le a < b < \infty$,

$$E_\varepsilon(b) + 2\int_a^b \int_M f_\varepsilon^{m+1}\, d\mu\, dt \le E_\varepsilon(a). \tag{15}$$

For every $t > 0$ they also satisfy

$$E_\varepsilon(t) \le C M_0^{\frac{N(m-1)+2s(m+1)}{N(m-1)+2s}}\, t^{-\frac{N-2s}{N(m-1)+2s}}, \tag{16}$$

where C is independent of ε.

Proof. By (10) and (14),

$$0 \le V_\varepsilon(t) \le C\varepsilon^{s-N/2} M_0, \qquad E_\varepsilon(t) \le C\varepsilon^{s-N/2} M_0^2. \tag{17}$$

Applying the positive heat kernel to (13) gives

$$V_\varepsilon(a) - V_\varepsilon(b) = \int_a^b P_\varepsilon(u(t)^m)\, dt \ge 0. \tag{18}$$

This positive integral is bounded by $V_\varepsilon(a)$; global integrability of u^m has not been used.

The mixed terms $\int f_\varepsilon(a) V_\varepsilon(b)$ and $\int f_\varepsilon(b) V_\varepsilon(a)$ are finite by (17) and the mass bound. They are equal by symmetry of G_s. Expanding the product and canceling them, then using positive Tonelli in (18), gives

$$\begin{aligned} E_\varepsilon(a) - E_\varepsilon(b) &= \int_M (V_\varepsilon(a) - V_\varepsilon(b))(f_\varepsilon(a) + f_\varepsilon(b))\, d\mu \\ &= \int_a^b \int_M P_\varepsilon(u(t)^m)(f_\varepsilon(a) + f_\varepsilon(b))\, d\mu\, dt. \end{aligned} \tag{19}$$

In particular E_ε is nonincreasing.

For fixed ε, x, the function $t \mapsto f_\varepsilon(t,x)$ is continuous. On a fixed ball this follows from local L^1 continuity and boundedness of $p_\varepsilon(x,\cdot)$. The contribution of its complement is uniformly small, since (3) bounds it by $2CM_0\varepsilon^{-N/2} e^{-R^2/(C\varepsilon)}$ outside $B_R(x)$. Sum (19) over partitions of $[a,b]$ whose mesh tends to zero. The endpoint multipliers converge pointwise to $2f_\varepsilon(t,x)$; all integrands are nonnegative. Fatou gives

$$E_\varepsilon(b) + 2\int_a^b \int_M P_\varepsilon(u^m) f_\varepsilon\, d\mu\, dt \le E_\varepsilon(a).$$

Conservativeness makes the heat kernel a probability measure, so Hölder gives $P_\varepsilon(u^m) \ge (P_\varepsilon u)^m$, also for unbounded inputs by monotone truncation. If $f_\varepsilon(t,x) = 0$, then $u(t) = 0$ almost everywhere for that kernel measure, so $P_\varepsilon(u(t)^m)(x) = 0$ as well. Thus the product is unambiguous and (15) follows.

Set $q = 2N/(N+2s)$ and $q' = 2N/(N-2s)$. These exponents are conjugate and $1 < q < N/(2s)$. Since $f_\varepsilon \in L^1 \cap L^\infty$, the estimate (7) and Hölder give

$$E_\varepsilon \le C \|f_\varepsilon\|_q^2 \le C M_0^\gamma \left(\int_M f_\varepsilon^{m+1} \, d\mu \right)^\beta,$$

where

$$\eta = \frac{(m+1)(N-2s)}{2Nm}, \qquad \beta = \frac{2\eta}{m+1} = \frac{N-2s}{Nm} \in (0,1), \qquad \gamma = 2(1-\eta) > 0.$$

Indeed $1/q = 1 - \eta + \eta/(m+1)$, which is the L^1–L^{m+1} interpolation identity. Consequently (15) implies

$$E_\varepsilon(b) + k_0 \int_a^b E_\varepsilon(t)^r \, dt \le E_\varepsilon(a), \qquad r = 1/\beta > 1, \quad k_0 = 2C^{-1/\beta} M_0^{-\gamma/\beta}. \tag{20}$$

Here a scalar comparison can be proved without an absolute continuity assumption. For any finite nonnegative nonincreasing E satisfying (20), monotonicity gives $k_0(b-a)E(b)^r \le E(a)$. If $E(t) > 0$, iterate this inequality on $[t/2^{j+1}, t/2^j]$ to obtain

$$\log E(t) \le r^{-n} \log E(0) + \sum_{j=0}^{n-1} r^{-(j+1)} [(j+1)\log 2 - \log(k_0 t)].$$

Letting $n \to \infty$ gives

$$E(t) \le 2^{r/(r-1)^2} (k_0 t)^{-1/(r-1)}.$$

If $E(t) = 0$, or $E(0) = 0$, the same bound is immediate. Apply it separately for each ε, using (17). Since

$$\frac{1}{r-1} = \frac{N-2s}{N(m-1)+2s}, \qquad \frac{\gamma}{1-\beta} = \frac{N(m-1)+2s(m+1)}{N(m-1)+2s},$$

we obtain (16), uniformly in ε. □

Proof of Theorem 2.2. Fix any $0 < a < b < \infty$. Proposition 4.1 bounds $\int_a^b \int_M f_\varepsilon^{m+1} \, d\mu \, dt$ uniformly in ε. The heat semigroup is strongly continuous on $L^1(M)$. One may see this directly from its L^2 continuity on bounded compact inputs, local L^1 convergence, and conservativeness, using the compact tightness argument in Proposition 3.2; density and contraction give the general case. Thus, for each t, $P_\varepsilon u(t) \to u(t)$ in L^1. The difference has norm at most $2M_0$, so dominated convergence in time yields convergence in $L^1((a,b) \times M)$. Take a subsequence with summable errors to obtain almost everywhere convergence. Fatou and the uniform dissipation bound imply

$$\int_a^b \int_M u^{m+1} \, d\mu \, dt < \infty.$$

Hölder in time and space, applied to $u^m = u^{1/m} (u^{m+1})^{(m-1)/m}$, now gives

$$\int_a^b \int_M u^m \, d\mu \, dt \le [(b-a)M_0]^{1/m} \left(\int_a^b \int_M u^{m+1} \, d\mu \, dt \right)^{(m-1)/m} < \infty.$$

In particular the potential difference $W_{a,b}$ in (13) belongs to $L^1(M)$. The equality assertion of Lemma 3.1, with $h = u(a) - u(b)$, gives $\int_M u(a) \, d\mu = \int_M u(b) \, d\mu$. All strictly positive-time masses therefore coincide. Finally let $a \downarrow 0$, using the strong initial L^1 continuity in Proposition 3.2. Their common value is $\int_M u_0 \, d\mu$. Since (13) holds at every endpoint, this proves the claimed equality for every finite $t > 0$. □

AI-use statement. OpenAI Codex was used to develop and check the arguments and to draft and revise the text, including internal reviews in fresh AI contexts. No machine-checked formalization or independent human verification is claimed.

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